

AI-Augmented Software Engineering Training

Program Overview

This training program prepares junior developers to work in modern software environments where **AI is an integral part of the development process**.

Participants learn how to design, build, test, and support software systems using a combination of traditional engineering practices and modern AI tools, agents, and context-aware systems. The program is aligned with real-world enterprise development processes and emphasizes practical, production-oriented skills.

Training Methodology

The training follows an end-to-end, project-driven approach that mirrors the full software development lifecycle (SDLC). Learning is structured around short theoretical introductions followed by extensive hands-on practice, guided exercises, and iterative feedback.

The methodology emphasizes:

- Learning by building real systems, not isolated exercises
- Continuous application of AI tools throughout the development process
- Incremental project development from requirements to production
- Collaboration, code reviews, and reflective learning



- Preparing participants for independent learning and adaptation in fast-changing environments

The program culminates in a final end-to-end project that simulates real-world software development in an AI-driven environment.

Training Specification

- Online lessons + hands-on experience
- Two projects and presentations – midterm and final
- Duration – 10–12 weeks
- ~180 hours training
- ~25 hours dedicated to projects and presentations

Core Technical Domains

- AI-Assisted Requirements Analysis & High-Level Design
- Python Software Engineering & Backend Development
- API Development & System Integration
- Git-Based Team Development & Code Review
- AI-Native IDE Workflow (Cursor / Copilot)
- Context Integration via MCP (Model Context Protocol)
- Agent Orchestration with LangChain / LangGraph
- LLM Applications: Prompting, RAG & Agents
- AI-Driven Testing, Debugging & Automation
- System Architecture for AI-Powered Applications
- AI-Enabled Production Monitoring & Support

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Syllabus

Module	Duration (Hours)	More
Course onboarding	5	Python environment setup; GitHub repo; Install Cursor/Copilot; Intro to AI workflow
Programming Foundations (Python)	30	Syntax, OOP, data structures, error handling, code quality
Software Development Lifecycle (SDLC)	10	Requirements → Design → Dev → Test → Deploy; AI in each phase
Source Control – Git	10	Git flow, branching, PRs, code reviews
Backend & API Development	25	REST APIs, data handling, system integration
AI-Native IDE Workflow	12	Cursor/Copilot, prompt-based coding, refactoring
LLM Applications	15	Prompting, RAG, intro to Agents
Agent Orchestration	10	LangChain / LangGraph, stateful workflows
Context Integration (MCP)	10	Connecting LLMs to tools, APIs, data
AI-Driven Testing & Automation	10	AI-generated tests, debugging
System Architecture	5	Designing AI-powered systems
Production Monitoring & Support	5	Logs, basic AI support agents
Final Project	25	End-to-end AI system + presentation

Total: 182 hours